

TEST BANK

CHAPTER 7

MEIOFAUNA

Matching

In this section, you are to select from the choices listed in column B the term or phrase that best matches the term or phrase listed in column A and then to place the letter for the selected choice from column B in the space provided to the left of the corresponding item in column A.

[Note to instructors: Here, the choices are correctly matched and thus should be randomly assorted before the exam is prepared.]

Column A		Column B
___ 1. permanent meiofauna	—————>	A. life cycle is restricted to the interstitial habitat
___ 2. elutriation	—————>	B. method to separate meiofauna
___ 3. statocysts	—————>	C. sensory organs that detect gravity
___ 4. spermatophore	—————>	D. harpacticoid copepods
___ 5. meiofaunal organisms	—————>	E. psammon
___ 6. thiobios	—————>	F. meiofauna of anoxic sediments
___ 7. temporary meiofauna	—————>	G. juveniles of macrofaunal animals
___ 8. vermiform	—————>	H. worm shaped
___ 9. MgCl ₂	—————>	I. an anesthetic agent
___ 10. species richness	—————>	J. number of different species in a habitat

Multiple Choice

Circle the choice that best completes or answers the given questions.

[N.B.: The correct answer to each of the questions is indicated by an asterisk. For most of the questions, the letter “a” is the correct choice; reorder the choices as you see fit.]

1. Light rarely penetrates below 5–15 _____ into the sediments.

a. μm
b. mm *
c. cm
d. m
e. km
2. Which one of the following phyla is not a member of the interstitial fauna?

a. Porifera *
b. Rotifera
c. Gastrotricha
d. Protozoa
e. Tardigrada

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3. Which one of the following phyla is not a member of the interstitial fauna?
 - a. Ctenophora *
 - b. Mollusca
 - c. Echinodermata
 - d. Brachiopoda
 - e. Arthropoda
4. Which one of the following phyla is not a member of the interstitial fauna?
 - a. Phoronida *
 - b. Bryozoa
 - c. Cnidaria
 - d. Kinorhyncha
 - e. Platyhelminthes
5. Endobenthic organisms are meiofaunal-sized animals that _____.
 - a. move by displacing the particles of the sediment that they inhabit *
 - b. move without displacing the particles of the sediment that they inhabit
 - c. live at the surface of the sediment–water interface
 - d. stay in the benthos for only a small part of their lives
6. Mesobenthic organisms are meiofaunal-sized animals that _____.
 - a. move without displacing the particles of the sediment that they inhabit *
 - b. move by displacing the particles of the sediment that they inhabit
 - c. live at the surface of the sediment–water interface
 - d. stay in the benthos for only a small part of their lives
7. Epibenthic organisms are meiofaunal-sized animals that _____.
 - a. live at the surface of the sediment–water interface *
 - b. move without displacing the particles of the sediment that they inhabit
 - c. move by displacing the particles of the sediment that they inhabit
 - d. stay in the benthos for only a small part of their lives
8. Besides water content, which one of the following is the most important environmental factor influencing the presence or absence of interstitial organisms?
 - a. Grain size *
 - b. Light
 - c. Salinity
 - d. Temperature
 - e. Oxygen concentration
9. Interstitial ciliated protozoans of the phylum Ciliophora tend to possess _____ body shapes.
 - a. elongate *
 - b. short and squat
 - c. spherical
 - d. amorphous
 - e. ringlike
10. Tardigrades possess which one of the following characteristics?
 - a. Four pairs of legs *
 - b. A swimming tail
 - c. An elongate proboscis
 - d. A set of long tentacles
 - e. A hard clamlike shell
11. In which one of the following seasons are interstitial organisms most abundant in the beaches of the northern hemisphere?
 - a. Summer *
 - b. Fall
 - c. Winter
 - d. Spring
12. The major interstitial groups have been found at water depths as great as _____ meters.
 - a. 5,000 *
 - b. 1,000
 - c. 500
 - d. 100
 - e. 10
13. Of the various geographic regions listed, which has been studied the least for its marine interstitial fauna?
 - a. Pacific coast of North America *
 - b. Atlantic coast of North America
 - c. Atlantic coast of Europe
 - d. Mediterranean Sea

14. Sediments in which the median particle size is $< 175 \mu\text{m}$ are dominated by which one of the following?
 - a. Burrowing organisms *
 - b. Forms that move between the sand grains
 - c. Forms that crawl over the top of the sediments
 - d. None of the above
15. The maximum number of meiofauna is generally found where the sand grains have a diameter of ____ mm.
 - a. between 0.175 and 0.275 *
 - b. < 0.1
 - c. > 0.5
 - d. > 1.0
16. Some meiofauna are termed eurythermal because they show _____.
 - a. considerable tolerance for a change in the temperature of their habitat *
 - b. no tolerance for a change in the temperature of their habitat
 - c. considerable tolerance for a change in the salinity of their habitat
 - d. no tolerance for a change in the salinity of their habitat
 - e. None of the above

True/False

Mark each of the given statements with the letter *T* if it is true and *F* if it is false.

[Instructors: In some cases, a few options are included for the same question.]

1. [True] Newly settled juveniles of macrofaunal organisms are called temporary meiofauna.
2. [False] Newly settled juveniles of macrofaunal organisms are called permanent meiofauna.
3. [False] Vertical migration is a behavior unknown to the meiobenthos.
4. [True] Statocysts are organs that detect gravity.
5. [False] No energy is transferred from the meiofauna to higher trophic levels.
6. [True] Several different zoogeographical regions or provinces are recognized based on the distribution of the meiofauna.
7. [False] All meiofauna have a cosmopolitan distribution; therefore, no zoogeographical regions or provinces are recognized.
8. [True] Meiofaunal organisms range in size from about 60 to 500 μm .
9. [False] Meiofaunal organisms range in size from about 500 to 1,500 μm .
10. [False] Most interstitial animals are sessile.
11. [True] Meiofauna have various ways to cling to sand grains.
12. [True] There is no latitudinal gradient in species richness in the meiofauna.
13. [True] Temperature changes over the course of a day are minimal in subtidal sediments.
14. [False] Temperature changes over the course of a day in the interstitial habitat below 15 cm are very great.
15. [True] A solution of MgCl_2 in seawater is a method used to anesthetize meiofauna.
16. [False] There is no vertical zonation of interstitial animals in sandy beaches.
17. [True] Different associations of meiofauna are found in sediments that are dominated by siliceous sands and those that are dominated by carbonate sands.
18. [False] There is no difference in the associations of meiofauna found in sediments dominated by siliceous sands and those dominated by carbonate sands.
19. [False] Meiofaunal organisms show no zonation patterns based on the grain size of the sediments that they inhabit.

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20. [True] Meiofaunal organisms show definite zonation patterns based on the grain size of the sediments that they inhabit.
21. [False] Sand with a coarse grain size is an ideal habitat because it drains water so well.
22. [True] The interstitial fauna tends to disappear when the median grain size falls below a diameter of 0.1 mm.
23. [False] Temperature is of no importance in determining the presence or absence of the meiofauna.
24. [True] Some meiofauna can survive being frozen in the sediments.

Short Answer (Fill in the Blank)

1. Newly settled juveniles of macrofaunal organisms are called temporary meiofauna.
2. Statocysts are organs that detect gravity.
3. Statocysts are organs that detect gravity.
4. The thiobios is a specialized faunal assemblage confined to the anoxic layers of the interstitial zone.
5. The thiobios is a specialized faunal assemblage confined to the anoxic layers of the interstitial zone.
6. The most important factor influencing interstitial organisms is grain size.
7. Light rarely penetrates below about 5–15 mm into sediments.
8. The size range for meiofauna is from 62 μm to about 500 μm (0.5 mm).
9. A packet of sperms in a single compact biological structure is called a spermatophore.
10. An efficient method of extracting meiofauna from sediment involves repeatedly stirring the sediment and letting the water overflow into a fine-meshed net for sieving. This method is called elutriation.
11. Of the phylum Annelida, the Classes Oligochaeta and Polychaeta are found in the marine interstitial zone.

Essay Discussion Questions

Develop complete answers to these questions.

1. Discuss three physiological or anatomical adaptations of the meiofauna.
2. To what do the terms interstitial, psammon, and meiofauna refer? What do the modifiers meso- and thalasso- mean?
3. What are the most important factors influencing the types of organisms present in the interstitial environment? Examine how grain size affects water drainage and oxygen content.
4. How does wave action influence the psammon? During what seasons of the year would wave action be most influential? Discuss.
5. How does light affect the psammon?
6. Explain the concept of ecological grain.
7. How many different phyla are represented in the interstitial environment? List five, and develop a table that compares and contrasts these organisms on the basis of food size, food-gathering strategies, and life history.
8. What sorts of problems would you expect to run into when attempting to extract meiofauna from core samples?
9. What is the size range for interstitial organisms, and how does the size of the sand grains in this habitat influence the meiofauna?
10. Why are meiofaunal organisms mainly vermiform? What is a worm?
11. What adaptive significance(s) does the spermatophore device serve in interstitial (psammic) copepods? Discuss.
12. What adaptations are found in psammic organisms for the protection of their embryos?
13. There are a number of large organisms that inhabit mud flats and sandy beaches. Which ones would be consumers of meiofauna? Construct a hypothetical food web from this information. *[Instructors may wish to provide information to aid the students or permit them to consult their notes and a handout.]*

14. What is unusual about the faunal community of the thalassios?
15. Discuss the types of respiration (metabolism) that organisms living in the thalassios use.
16. How is wave action an important abiotic factor to the meiofauna?
17. Discuss the special adaptations that the meiofauna possess to cope with the risk of being thrown into the open water by wave action.
18. How are statocysts helpful to the meiofauna? Discuss.
19. What are the consequences of small grain size vs. large grain size to meiofauna?
20. Why is a freshwater flood usually of little matter to marine meiofauna? Discuss.
21. Consider two populations of meiofauna that are both 4 m from the water's edge. One is at a depth of 10 cm, and the other is at a depth of 50 cm. Using Figure 7.1, which illustrates a vertical transect of a sandy beach, compare and contrast the difficulties that each population would encounter.
22. Is the interstitial habitat uniform in all dimensions? Discuss.
23. Challenge Question: Gills and gill-like structures are essentially absent from the meiofauna. Discuss.
24. Challenge Question: Invent a completely new meiofaunal organism, and discuss its general Bauplan.
25. Challenge Question: Discuss how one might apply the concept of the microbial loop in the marine pelagic zone to the interstitial habitat. [N.B.: The microbial loop is considered in Chapter 2.]
26. Challenge Question: In what ways might terrestrial soils be similar to the marine interstitial environment? In what ways might they be different?
27. Challenge Question: In what ways might freshwater sediments be similar to the marine interstitial environment? In what ways might they be different?
28. Challenge Question: The biomass ratio of meiofauna to macrofauna falls in the range of 1:5–1:10; yet in intertidal sands and mud flats and in the deep sea, this ratio approaches 1:1. Discuss this phenomenon. [*Hint*: Consider the trophic dynamics of these two habitats.]
29. Speculate as to why meiofaunal organisms are distributed in patches.

